EXPERIMENT E7 · VARIANT C · MEASUREMENT REVIEW → REVISION B (FRAME REV. B3)

# Will the deformation measurement work?

27 Oct 2026 · read as the person who will run the tests · numbers from the axisymmetric heat model (sim_thaw.py), indicative

**Verdict.** The rig will run, and temperature, TDR and Pt100 will be measured without trouble. Deformation is the delicate part: in the unsaturated sand (Sr = 0.5, decided 9 Nov) the expected heave is about as large as the thermal drift of the rig itself, so E7 measures mainly the thaw settlement under load. Revision B removes most of the rig-side problems.

## Findings and what revision B changes

| TOPIC | PROBLEM IN REV. A | CHANGE IN REV. B | EFFECT | STATUS |
|---|---|---|---|---|
| **Signal size** | Sr = 0.5: ice adds 1.7 % volume, pore air is 19 % → in a closed sample the expansion fits into the pores | none on the rig; Sr 0.5 chosen (see below) | expected in-situ heave ≈ 0 (tens of µm at most); upper bound 1.7 mm | ✓ decided |
| **Rig drift** | LVDT on the Al beam with a steel rod; camera referenced to the PETG rim | Invar 36 post from the base frame carries the LVDT and a camera target; Invar guide rod | per 20 K: LVDT −24…+48 → −59…0 µm, left after blank-run correction 12 → ≤ 6 µm; camera 0…100 → −49…0 µm, left 8 → ≤ 3 µm | ✓ fixed |
| **Freezing direction** | cell on a 25 mm Al plate freezes from below and the sides too; last unfrozen spot 35 mm above the bottom | cell printed in one piece ⌀240 × 200: +40 mm low-infill wall and base (λ ≈ 0.06), 3 G10 pads in base pockets; frame of T-slot profiles, housing under the top beam, beam gap 140 → rig 559 mm high | freezes from the top: last unfrozen 13 mm above the bottom; 95 % frozen in 15 h (was 10.5 h) | ✓ fixed |
| **Laser** | 45° laser: collar shadows a 25 mm strip, 41° DOE gives only 7–8 lines | laser at 68° on a bracket under the beam, 124 mm from the centre | no shadow, 11 lines at 9.6 mm; line shift 0.40 Δh | ✓ fixed |
| **Camera** | CSI ribbon cannot pass the lid seal; separate heater near the sample | RPi 5 inside the camera box (its ≈ 5 W keep the window clear); the RPi takes photos on its NTP-synced clock; 6 mm lens | Ethernet + 5 V through the seal only; time stamps within < 1 s of the CR850 | ✓ fixed |
| **Bearing** | LM12UU ball bushing: grease stiffens at −10 °C, ice at thaw → stick-slip, load not constant | igus drylin RJ4JP-01-12, dry polymer, same size | load change from friction < 0.02 kPa | ✓ fixed |
| **Moisture** | open surface sublimates while frozen; frost and dew at the start of thaw look like heave | witness dish with the same sand weighed each cycle (the cell stays in place, weighed only at the start and end of a series); SHT45 air RH on the RPi | loss and frost become visible in the data | ✓ added |
| **Freezer air** | on/off control swings ±2–3 K; air sensor only optional | 80 mm fan; Campbell 109 air probe required | boundary condition known; smaller drift noise | ✓ added |

### ✓ Decided 9 Nov: Sr = 0.5

E7 measures the **thaw settlement under load** in unsaturated sand.

- **Expect tens of µm** per cycle; rely on the blank-run correction (LVDT ≤ 6 µm, camera ≤ 3 µm left).
- **Frost heave** would need Sr ≥ 0.85 or water supplied from below – not part of E7.

### Before the first real cycle

1. Blank run: Al dummy instead of sand, one full cycle; fit drift against cell temperature.
2. Calibrate the camera with shims 0.1 / 0.5 / 1.0 mm under a flat plate.
3. Measure the freezer: rig 559 mm vs ≈ 650 mm inside.
4. Confirm LVDT operation at −10 °C with RDP / Elkome.
5. Program: 24 h at −10 °C, 24 h at +10 °C (thaw takes ≈ 15 h).

## Cost and resolution

**≈ 1,460–1,990 €**
rev. B excl. VAT (rev. A 1,365–1,870 €); frame from T-slot profiles cut to length and printed parts, no machining

**≤ 6 µm**
LVDT drift left after the blank-run correction (model), LVDT resolution ≈ 1 µm

**≈ 10 µm**
camera height map per laser line after averaging ≈ 1000 px along the line

Details: drawings E7-V01…V05 rev. B, purchase list, calculations §15 (vypocty.md), 3D model with drift chart: E7 sample thawing.

Model assumptions: Sr 0.5, θ 0.19, n 0.38, λ sand 1.3 / 2.0 W/mK, h = 7 W/m²K in still air; frame parts as lumped elements. Treat all numbers as estimates to be checked by the blank run.